

Algorithms: Assignment 1

1. Show a comparison based algorithm for finding both the minimum *and* maximum in an unsorted array of n elements using $\lceil 3n/2 \rceil - 2$ comparisons. Also show that $\lceil 3n/2 \rceil - 2$ comparisons are necessary to find the minimum and maximum.
2. Show that the minimum and second minimum elements in an unsorted array of n elements can be determined using $n + \log n - 2$ comparisons. Assume n to be a power of 2.
3. Given a set S of n distinct numbers and a positive integer $k \leq n$, show an $O(n)$ algorithm to determine the k numbers in S that are closest to the median of S .
4. Let $X[1..n]$ and $Y[1..n]$ be two arrays, each containing n numbers already in sorted order. Give an $O(\log n)$ -time algorithm to find the median of all $2n$ numbers in arrays X and Y .
5. For n distinct elements $w_1, \dots, w_n$ such that $\sum_{i=1}^n w_i = 1$, their *weighted (lower) median* is the element w_k satisfying

$$\sum_{w_i < w_k} w_i < \frac{1}{2} \quad \text{and} \quad \sum_{w_i \geq w_k} w_i \leq \frac{1}{2}.$$

For example, if the elements are 0.1, 0.35, 0.05, 0.1, 0.15, 0.05, 0.2, then the weighted median is 0.2. Show how to compute the weighted median of n distinct elements in $O(n)$ time.

6. Given a list of values $z_0, z_1, \dots, z_{n-1}$ (possibly with repetitions), show how to find the coefficients of a polynomial $P(x)$ of degree n that has zeroes only at $z_0, z_1, \dots, z_{n-1}$ in time $O(n \log^2 n)$.
7. Consider two sets A and B , each having n integers in the range from 0 to $10n$. We wish to compute the *Cartesian sum* of A and B , defined by

$$C = \{x + y : x \in A, y \in B\}.$$

Note that the integers in C are in the range 0 to $20n$. We want to find the elements in C and the number of times each element of C is realized as a sum of elements in A and B . Show that the problem can be solved in $O(n \log n)$ time.

8. We are given 2 positive integers $\langle a_{n-1}a_{n-2} \dots a_0 \rangle$ and $\langle b_{n-1}b_{n-2} \dots b_0 \rangle$, of n bits each, as 2 arrays $A[1..n]$ and $B[1..n]$, respectively. The MSB of a , that is a_{n-1} , is stored in the location $A[1]$ and so on. Similarly with b . Give an $O(n \log n)$ algorithm to compute the array $C[1..2n]$ that holds the integer ab (the product of a and b).
9. Show that the number of distinct minimum cuts in an undirected graph (assume all edge weights are one) is at most $\binom{n}{2}$. That is, show that the number of distinct cuts whose value is equal to the value of the min cut in the graph is at most $\frac{n(n-1)}{2}$.
10. Suppose that at each step of the simple contraction algorithm for min-cut, instead of choosing a random edge for contraction, we choose two vertices at random and merge the two vertices into a single vertex. Show that there are inputs on which the probability that this modified algorithm finds a min-cut is exponentially small.